01_POWER_MEASUREMENT

POWER BUDGET: total load from STLINK_5V must remain <= 200 mA

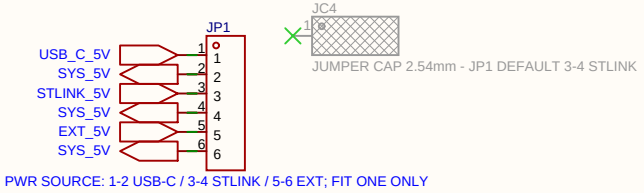
Station-S Rev0: power entry, 3.3 V generation, DUT current/voltage measurement

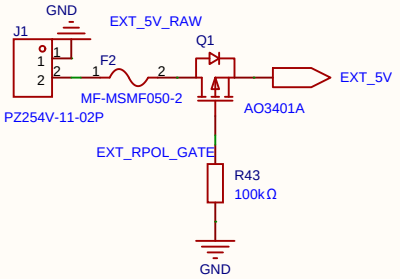
TEST POINTS

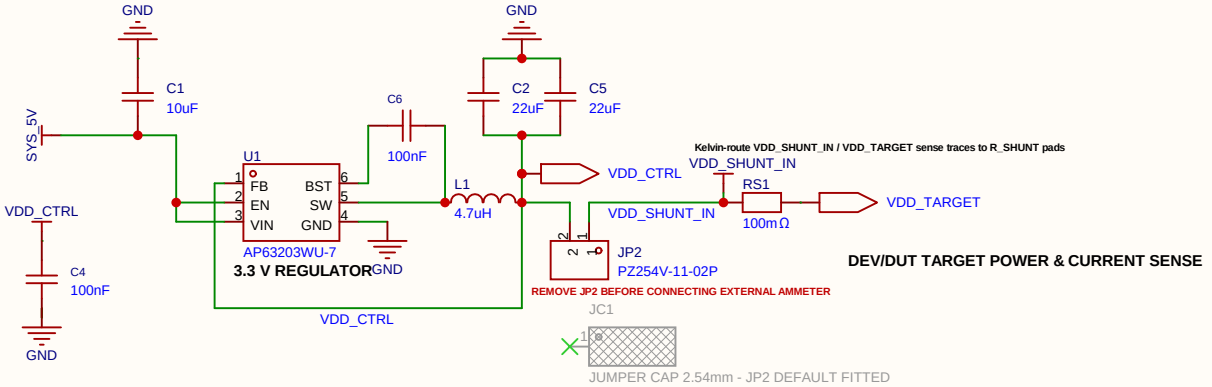
TP1 STLINK_5V
TP2 SYS_5V
TP3 VDD_CTRL
TP4 VDD_SHUNT_IN
TP5 VDD_TARGET
TP6 GND
TP7 MEAS_SCL
TP8 MEAS_SDA
TP9 MEAS_ALERT

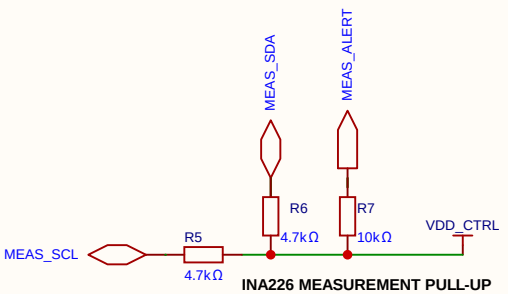
POWER INPUT / SELECTION

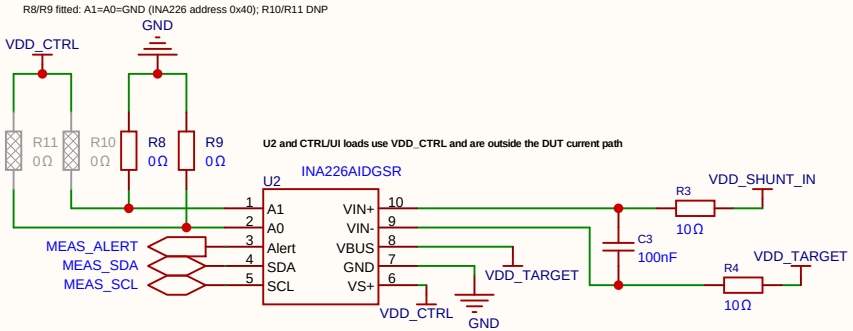
J1: optional EXT_5V input; JP1 default jumper 3-4 selects STLINK_5V





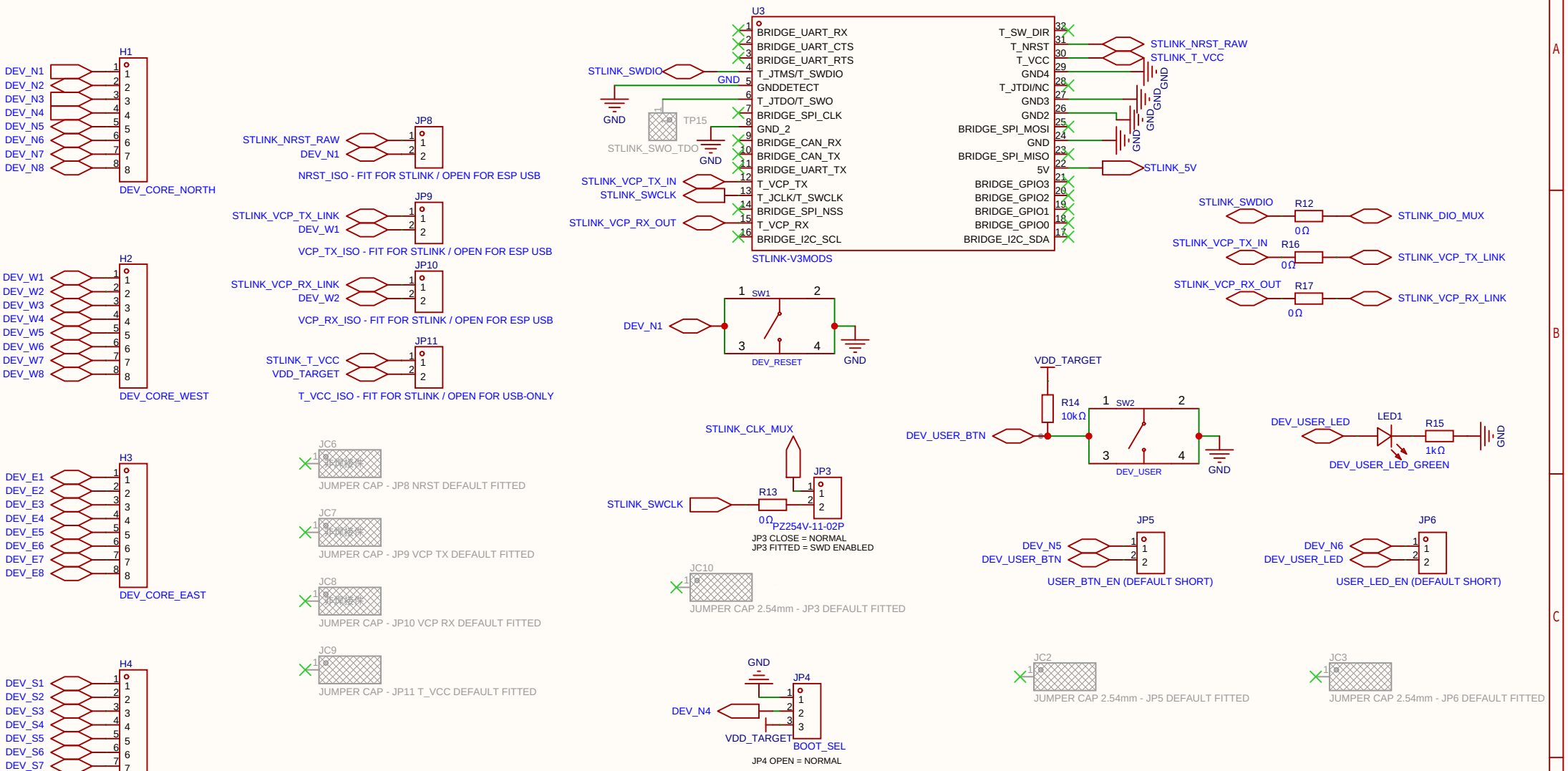






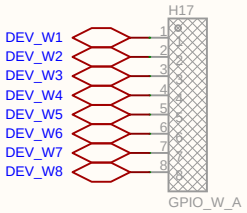
Schematic	Schematic1	Create at	2026-09-15
		Update at	2026-09-28
Board	Board1	Page	01_PWR_MEAS
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version	Size
EasyEDA	V1.0	A4	Page 1 Total 7
			CraftVia.org

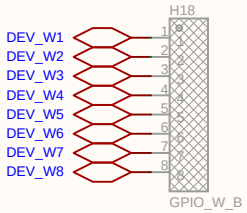
02_DEV_DEBUG

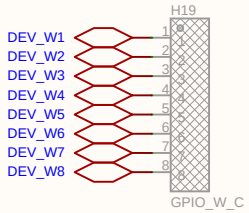


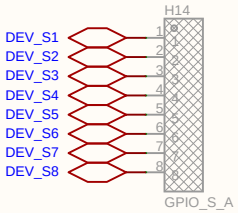
Schematic	Schematic1	Create at	2026-09-15
		Update at	2026-09-30
Board	Board1	Page	02_DEV_DEBUG
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version	Size	Page 2 Total 7
EasyEDA		V1.0	A4	CraftVia.org

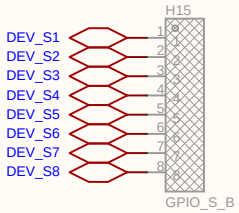
03_GPIO_BREAKOUT

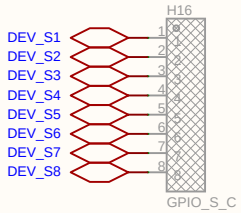


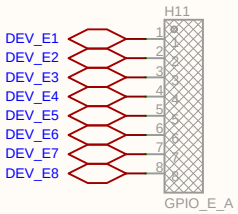


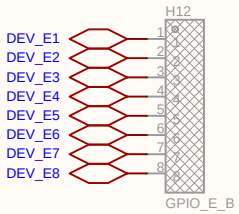


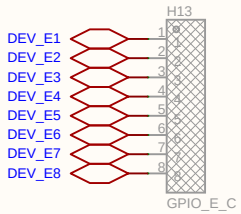


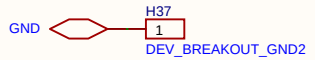


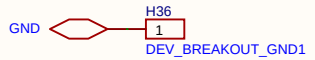


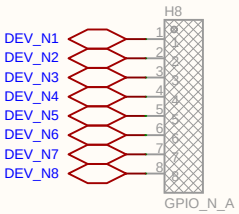


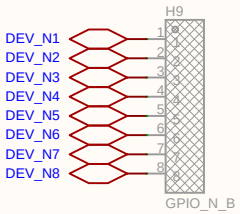


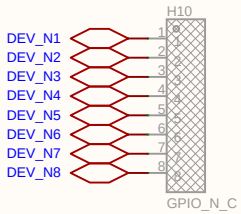


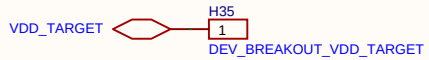






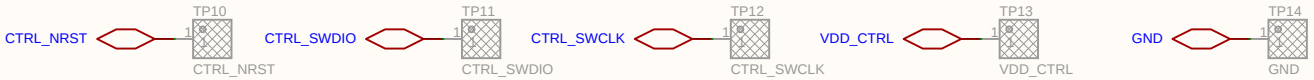




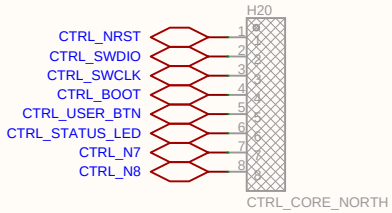


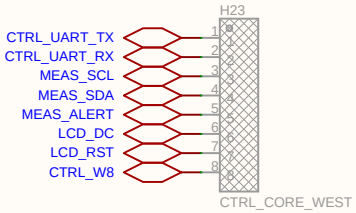
Schematic	Schematic1	Create at	2026-09-16
		Update at	2026-09-28
Board	Board1	Page	03_GPIO_BREAKOUT
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version	Size	Page 3 Total 7
EasyEDA		V1.0	A4	CraftVia.org

04_CTRL_CORE



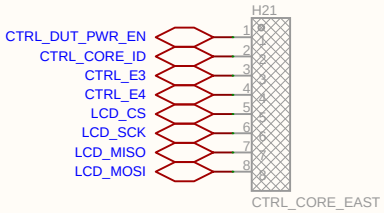
Recovery / fault-analysis test points

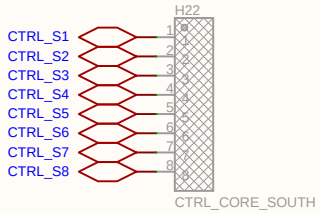














Program the Core-S in DEV/DUT socket, then move it here.

CTRL Core-S is not connected to STLINK-V3MODS SWD.

CTRL Core-S socket / measurement and UI control

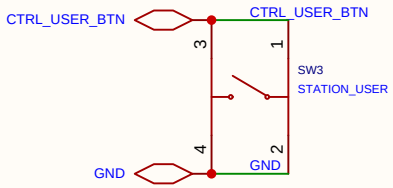
Schematic	Schematic1	Create at	2026-07-26
		Update at	2026-09-25
Board	Board1	Page	04_CTRL_CORE
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version	Size	Page 4 Total 7
EasyEDA		V1.0	A4	CraftVia.org

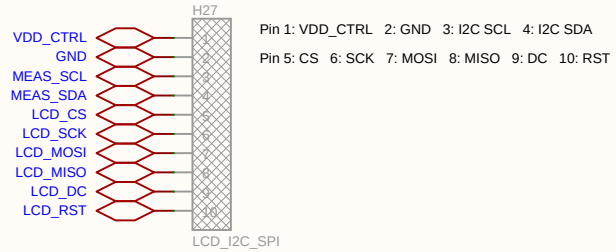
05_CTRL_UI



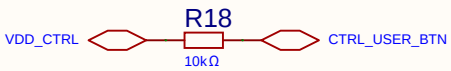
STATUS LED (Active High)

Station control user interface





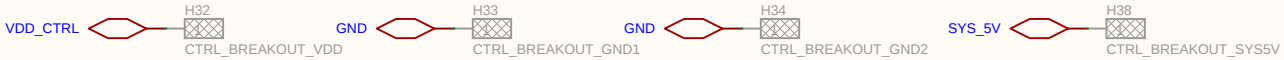
LCD connector: I2C or SPI, removable



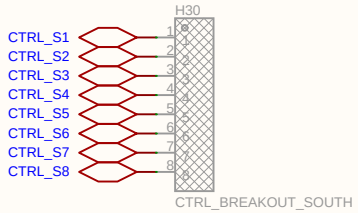
STATION USER button (Active Low)

Schematic	Schematic1	Create at	2026-09-16
		Update at	2026-09-28
Board	Board1	Page	05_CTRL_UI
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
	Version	Size	Page 5 Total 7
EasyEDA	V1.0	A4	CraftVia.org

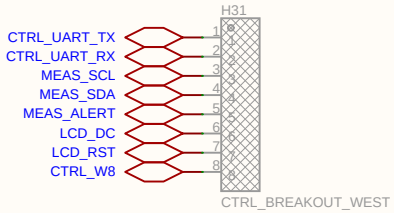
06_CTRL_BREAKOUT



Power / GND access



SOUTH S1 – S8



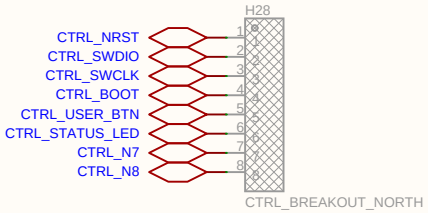
WEST W1 – W8

Final size decided during PCB placement

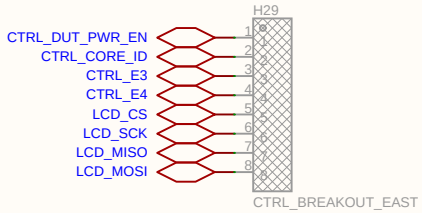
No default copper or power connection

2.54 mm isolated through-hole grid

PCB ONLY — UNIVERSAL AREA



NORTH N1 – N8

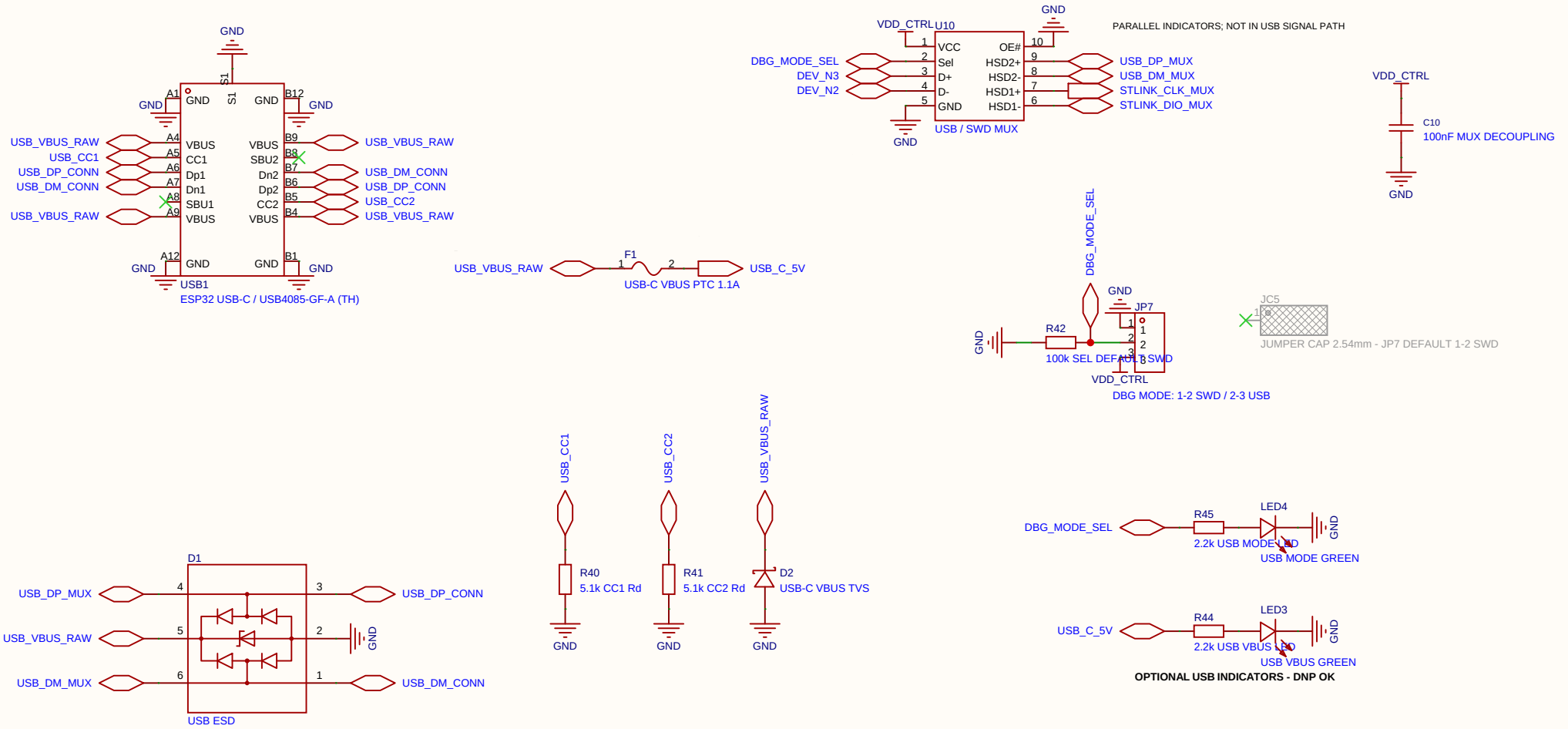


EAST E1 – E8

CTRL Core-S breakout — one 2.54 mm through-hole per terminal

Schematic	Schematic1	Create at	2026-09-16
		Update at	2026-09-28
Board	Board1	Page	06_CTRL_BREAKOUT
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version	Size	Page 6 Total 7
EasyEDA		V1.0	A4	CraftVia.org

07_USB_ESP



Schematic	Schematic1	Create at	2026-09-25
		Update at	2026-09-28
Board	Board1	Page	07_USB_ESP
Drawn		CraftVia-Station-S-v1-Rev0	
Reviewed			
		Version / Size / Page 7 Total 7	
EasyEDA	V1.0 / A4	CraftVia.org	